PROBLEM 43

Two Connected Particles through a Hole

Original statement · Problems.pdf, p. 30

Angular Momentum + Energy

We consider a smooth horizontal plane wih [sic] a small hole in it. One of the ends of a massless cord of length $2a$ is passed through the hole and then, two point particles, P_1 and P_2, each with the same mass m, are attached to the ends of the cord. Therefore, P_1 is constrained to move on the plane and P_2 moves along a vertical line below the hole.

Initially, half of the cord lies under the plane and half of the cord is on the plane. At that instant, P_1 is pushed with a speed v_0 perpendicular to the cord, counterclockwise as seen from above. Assume that the cord is always in tension.

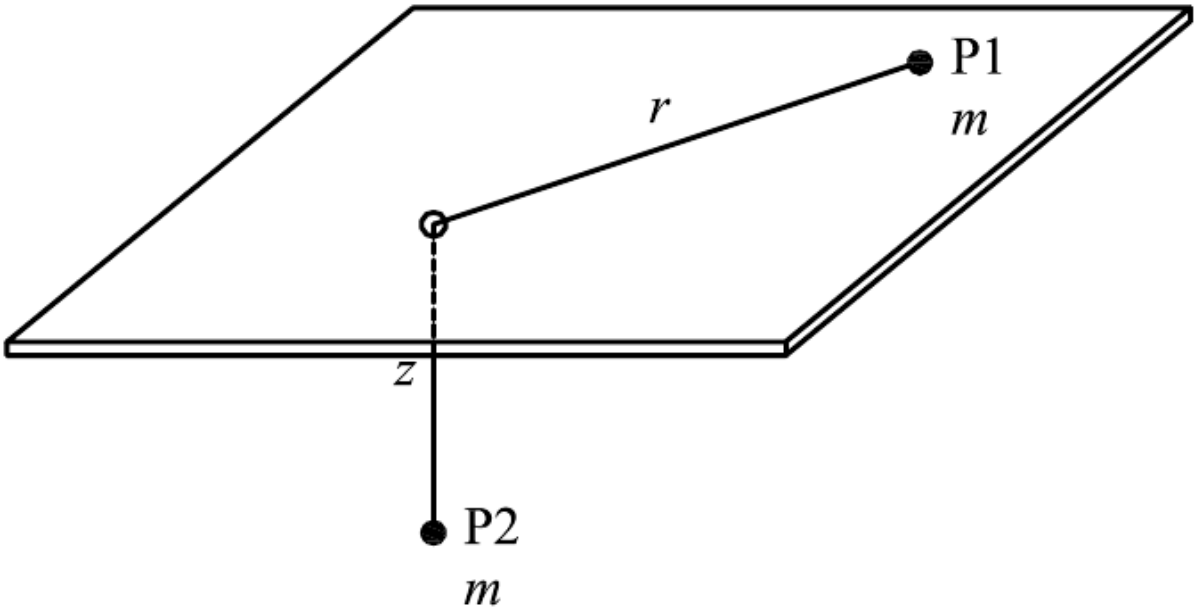


Figure: Smooth horizontal plane with a small hole, particle P1 on the plane at distance r from the hole and particle P2 hanging below it, both of mass m (Problems.pdf, p. 30).

(a) What is the value of v_0 if P_1 describes a circumference?

(b) In that case, what is the value of the tension in the cord?

(c) If $v_0 = \sqrt{8ag/3}$, reduce the problem to integrals (i.e., quadratures).

(d) Show that for this value of v_0, P_1 oscillates between $r = a$ and $r = 2a$.

(e) Apply Newton's law to P_2 and show that the tension of the cord is $T = mg\left(\frac{1}{2} + \frac{4a^3}{3r^3}\right)$.

(f) Write down the new equations of motion if the plane on which P_1 moves is rough, with a friction coefficient μ.

Hint: the roots of the polynomial $-3x^3 + 7x^2 - 4$ are $x_1 = 1$, $x_2 = 2$ and $x_3 = -2/3$.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** P_1 on a smooth horizontal plane at polar position (r, θ) and P_2 hanging on the vertical through the hole O, joined by an inextensible massless cord of length $2a$ that passes through O.
- **Hypotheses:** inertial frame S_0 with O at the hole; smooth hole and plane for parts (a) to (e); the cord stays taut ($T > 0$, checked in (e)); no drag.
- **Constraints** (holonomic and scleronomic; the cord is unilateral in principle, so $T \geq 0$ is verified afterwards):
  - P_1 on the plane: $z_1 = 0$.
  - P_2 on the vertical through the hole: $x_2 = y_2 = 0$.
  - Cord of constant length: $r + \ell_2 = 2a$, where ℓ_2 is the hanging length, so $z_2 = -\ell_2 = r - 2a$ (Oz pointing up).
  - Geometric feasibility: $\ell_2 \geq 0$, i.e. $r \leq 2a$ (P_2 cannot go above the hole). This is a unilateral limit that the solution must respect (see Phase 4).

- **Degrees of freedom:** two particles have 6 coordinates and there are 4 independent constraints, so $\mathrm{CDOF} = 6 - 4 = 2$.
- **Coordinates:** (r, θ), polar coordinates of P_1 on the plane.
- **Initial state:** $r(0) = a, \dot{r}(0) = 0, r(0)\dot{\theta}(0) = v_0$, hence $\dot{\theta}(0) = v_0/a$ and $\dot{z}_2(0) = 0$.

Symbol	Meaning	SI unit
m	mass of each particle	kg
a	half of the cord length	m
v_0	initial speed of P_1	m/s
g	gravitational acceleration	m/s^2
μ	friction coefficient (part f)	1

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, horizontal plane $z = 0$, $\mathbf{k}$ up.
- **Polar basis of P_1.** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1]^T[{}_0R_1] = I$$

Multiplying out, RR^T has diagonal entries $\cos^2 + \sin^2 = 1$ of the rotation angle (and 1) and off-diagonal entries $\pm(\cos\sin - \sin\cos) = 0$, so it is the identity.

- $\boldsymbol{\omega}_{10} = \dot{\theta}\,\mathbf{k}$, so by Poisson $\dot{\mathbf{e}}_r = \dot{\theta}\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot{\theta}\,\mathbf{e}_r$. P_2 moves along $\mathbf{k}$, so it needs no extra basis.

PHASE 3

Derivation

3.1 Kinematics

Pedagogical Justification: the constraints leave r and θ free; the position of P_2 follows from the cord constraint $z_2 = r - 2a$. For P_1 we use the polar formulas of Problem 42 (product rule and Poisson).

$$\mathbf{r}_1 = r\mathbf{e}_r, \qquad \mathbf{v}_1 = \dot{r}\,\mathbf{e}_r + r\dot{\theta}\,\mathbf{e}_\theta, \qquad \mathbf{a}_1 = \left(\ddot{r} - r\dot{\theta}^2\right)\mathbf{e}_r + \left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right)\mathbf{e}_\theta$$

$$\mathbf{r}_2 = (r - 2a)\,\mathbf{k}, \qquad \mathbf{v}_2 = \dot{r}\,\mathbf{k}, \qquad \mathbf{a}_2 = \ddot{r}\,\mathbf{k}$$

3.2 Forces

- **On P_1:** its weight $-mg\,\mathbf{k}$; the normal reaction of the smooth plane $N_1\,\mathbf{k}$ (unknown magnitude, vertical because the plane is smooth); and the tension of the cord $-T\,\mathbf{e}_r$ (the cord is horizontal on the plane and pulls toward the hole).
- **On P_2:** its weight $-mg\,\mathbf{k}$, the tension $+T\,\mathbf{k}$ (same cord, massless, smooth hole) and the horizontal reaction of the vertical line $R_{2x}\,\mathbf{i} + R_{2y}\,\mathbf{j}$ that keeps $x_2 = y_2 = 0$ (unknown, to be determined).
- **Inertia forces:** none, because the frame S_0 is inertial.
- **Unknowns:** $r(t), \theta(t), T(t), N_1(t), R_{2x}(t)$ and $R_{2y}(t)$.

3.3 Equations of Motion

> **Pedagogical Justification:** Newton's second law, one vector equation for each particle, gives six scalar equations. The vertical equation of P_1 and the horizontal ones of P_2 only fix reactions, so we project on the tangent directions of the constraints first.

$$P_1,\ \mathbf{e}_r:\ m\left(\ddot{r} - r\dot{\theta}^2\right) = -T, \qquad P_1,\ \mathbf{e}_\theta:\ m\left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right) = 0, \qquad P_1,\ \mathbf{k}:\ 0 = N_1 - mg$$

$$P_2,\ \mathbf{k}:\ m\ddot{r} = T - mg, \qquad P_2,\ \mathbf{i}:\ 0 = R_{2x}, \qquad P_2,\ \mathbf{j}:\ 0 = R_{2y}$$

- **Check of closure:** six unknowns ($r, \theta, T, N_1, R_{2x}, R_{2y}$) and six equations. The reactions are decoupled: $N_1 = mg$ and $R_{2x} = R_{2y} = 0$, because $\mathbf{a}_2$ is parallel to $\mathbf{k}$. Only r, θ and T remain, with three equations.
- **Dynamical decoupling.** Adding the $\mathbf{e}_r$ equation of P_1 and the equation of P_2 eliminates T:

$$m\left(2\ddot{r} - r\dot{\theta}^2\right) = -mg$$

- **Angular momentum.** Multiplying the transverse equation by r/m gives $\frac{d}{dt}\left(r^2\dot{\theta}\right) = r\left(r\ddot{\theta} + 2\dot{r}\dot{\theta}\right) = 0$, so $h = r^2\dot{\theta}$ is constant.
- **Energy.** The tension does work $-T\dot{r}$ on P_1 and $+T\dot{r}$ on P_2, which cancel (inextensible cord); the plane does no work. With the kinetic energy $K = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) + \frac{1}{2}m\dot{r}^2$ and $V = mgz_2 = mg(r - 2a)$, the energy is conserved. Replacing $r^2\dot{\theta}^2 = h^2/r^2$:

$$E = K + V = m\dot{r}^2 + \frac{mh^2}{2r^2} + mg(r - 2a) = \text{const}$$

3.4 Integration and Answers

(a) Circular motion, $r \equiv a$

> **Pedagogical Justification:** for r constant, $\dot{r} = \ddot{r} = 0$, so P_2 is at rest and the cord tension must balance its weight, while P_1 moves on a circle under the tension alone. With $\dot{r} = 0$ the transverse equation gives $r\ddot{\theta} = 0$, so $\dot{\theta}$ is constant and equal to its initial value v_0/a (equivalently $h = a^2\dot{\theta} = av_0$).

$$P_1:\ m\left(0 - a\dot{\theta}^2\right) = -T \implies T = ma\dot{\theta}^2 = \frac{mv_0^2}{a}, \qquad P_2:\ 0 = T - mg \implies T = mg$$

$$\frac{mv_0^2}{a} = mg \implies \boxed{v_0 = \sqrt{ag}}$$

(b) Tension in the circular case

$$\boxed{T = mg}$$

(c) Quadratures for $v_0 = \sqrt{8ag/3}$

> **Pedagogical Justification:** the two first integrals of 3.3 leave a single equation for $r(t)$. Evaluate h and E at the initial instant.

$$h = r^2\dot\theta = a v_0 = a\sqrt{\frac{8ag}{3}} \implies h^2 = \frac{8}{3}a^3 g, \qquad E(0) = \tfrac{1}{2}mv_0^2 + mg(a-2a) = \tfrac{1}{2}m\cdot\frac{8ag}{3} - mga = \frac{1}{3}mga$$

- Equating $E = E(0)$, dividing by m and isolating $\dot r^2$:

$$\dot r^2 + \frac{4a^3 g}{3r^2} + gr - 2ga = \frac{1}{3}ga \implies \dot r^2 = \frac{7}{3}ga - gr - \frac{4a^3 g}{3r^2} = \frac{g}{3r^2}\left(7ar^2 - 3r^3 - 4a^3\right)$$

- Separating variables, with $P(r) \equiv -3r^3 + 7ar^2 - 4a^3$. The sign of $\dot r$ is $+$ in the first half-cycle, because P_1 starts at the inner turning point $r = a$ with $\ddot r(0) = -\frac{g}{2} + \frac{h^2}{2a^3} = \frac{5g}{6} > 0$ (from the radial equation of 3.3 with $\dot\theta = h/r^2$). Applying Barrow's rule from $r(0) = a$ on the left side and from 0 to t on the right side:

$$\frac{dr}{dt} = +\sqrt{\frac{g}{3}}\,\frac{\sqrt{P(r)}}{r} \implies \int_a^r \frac{r'\,dr'}{\sqrt{P(r')}} = \sqrt{\frac{g}{3}}\int_0^t dt' = \sqrt{\frac{g}{3}}\,t, \qquad 0 \le t \le \frac{\tau_r}{2}$$

The integrand is improper at $r' = a$ (the radicand vanishes there) but integrable, since $P(r')$ has a simple root at a. After the outer turning point $r = 2a$, reached at $t = \tau_r/2$, the sign of $\dot r$ is $-$ and the quadrature restarts from $2a$:

$$\int_r^{2a} \frac{r'\,dr'}{\sqrt{P(r')}} = \sqrt{\frac{g}{3}}\left(t - \frac{\tau_r}{2}\right), \qquad \frac{\tau_r}{2} \le t \le \tau_r$$

and the motion repeats with the radial period τ_r (see Phase 4).

- **Quadrature for the angle.** With $\dot\theta = h/r^2$ and the same separation of variables, $d\theta = \dfrac{h}{r^2\dot r}\,dr$, and using $h/\sqrt{g/3} = \sqrt{8a^3}$. In the first half-cycle ($\theta(0) = \theta_0$) and in the second one:

$$\theta(r) = \theta_0 + \sqrt{8a^3}\int_a^r \frac{dr'}{r'\sqrt{P(r')}}, \qquad \theta(r) = \theta(2a) + \sqrt{8a^3}\int_r^{2a} \frac{dr'}{r'\sqrt{P(r')}}$$

or, directly in time, $\theta(t) = \theta_0 + \displaystyle\int_0^t \frac{h}{r(t')^2}\,dt'$, which is valid for all t because $\dot\theta = h/r^2 > 0$.

(d) Oscillation between $r = a$ and $r = 2a$

> **Pedagogical Justification:** motion requires $\dot r^2 \geq 0$, i.e. $P(r) = -3r^3 + 7ar^2 - 4a^3 \geq 0$. With $x = r/a$ the polynomial is $a^3(-3x^3 + 7x^2 - 4)$, and the hint gives its roots.

$$-3x^3 + 7x^2 - 4 = -(x-1)(x-2)(3x+2)$$

(check: $(x-1)(x-2) = x^2 - 3x + 2$ and $(x^2 - 3x + 2)(3x + 2) = 3x^3 - 7x^2 + 4$).

- Check of the roots: $x = 1$: $-3 + 7 - 4 = 0$; $x = 2$: $-24 + 28 - 4 = 0$; $x = -2/3$: $\frac{8}{9} + \frac{28}{9} - 4 = 0$. Since $r > 0$, $3x + 2 > 0$ and $P \geq 0$ requires $(x-1)(x-2) \leq 0$:

$$1 \leq x \leq 2 \iff a \leq r(t) \leq 2a$$

- The turning points ($\dot r = 0$) are $r_{\min} = a$ and $r_{\max} = 2a$. Both are simple roots of P, and the radial acceleration points back into the allowed interval at each one: $\ddot r(a) = \frac{5g}{6} > 0$ and $\ddot r(2a) = -\frac{g}{2} + \frac{h^2}{16a^3} = -\frac{g}{3} < 0$ (from $\ddot r = -\frac{g}{2} + \frac{h^2}{2r^3}$, derived in (e)). So r reverses in finite time at both ends and the motion oscillates between them.
- Feasibility: $r_{\max} = 2a$ is exactly the limit $\ell_2 = 2a - r \geq 0$ of Phase 1; at that instant P_2 reaches the hole with zero velocity.

(e) Tension of the cord

> **Pedagogical Justification:** the equation of P_2 from 3.3 gives $T = m\,(g + \ddot{r})$. We need $\ddot{r}$ as a function of r. It comes directly from the decoupled equation $2\ddot{r} - r\dot{\theta}^2 = -g$ of 3.3, with $\dot{\theta} = h/r^2$ (no division by $\dot{r}$ is needed):

$$2\ddot{r} = -g + \frac{h^2}{r^3} \implies \ddot{r} = -\frac{g}{2} + \frac{h^2}{2r^3} = -\frac{g}{2} + \frac{4a^3 g}{3r^3} \qquad \left(h^2 = \tfrac{8}{3}a^3 g\right)$$

- Check by differentiating the energy integral of (c) (chain rule, $\frac{d}{dt} r^{-2} = -2r^{-3}\dot{r}$): $2\dot{r}\ddot{r} = \left(-g + \dfrac{8a^3 g}{3r^3}\right)\dot{r}$, which gives the same $\ddot{r}$ wherever $\dot{r} \neq 0$.

$$\boxed{T = mg\left(\frac{1}{2} + \frac{4a^3}{3r^3}\right)}$$

- $T > 0$ for all $r \in [a, 2a]$, so the cord stays taut.

(f) Rough plane

> **Pedagogical Justification:** Coulomb friction acts on P_1 opposite to its velocity, with magnitude μN_1. The normal force is $N_1 = mg$ (3.3), because the cord is horizontal on the plane and only the weight acts vertically.

$$\mathbf{F}_f = -\mu mg\,\frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = -\mu mg\,\frac{\dot{r}\,\mathbf{e}_r + r\dot{\theta}\,\mathbf{e}_\theta}{\sqrt{\dot{r}^2 + r^2\dot{\theta}^2}}$$

$$\mathbf{e}_r:\ m\left(\ddot{r} - r\dot{\theta}^2\right) = -T - \mu mg\frac{\dot{r}}{\sqrt{\dot{r}^2 + r^2\dot{\theta}^2}}$$

$$\mathbf{e}_\theta:\ m\left(r\ddot{\theta}+2\dot{r}\dot{\theta}\right)=-\mu mg\frac{r\dot{\theta}}{\sqrt{\dot{r}^2+r^2\dot{\theta}^2}},\qquad P_2:\ m\ddot{r}=T-mg$$

- Adding the $\mathbf{e}_r$ equation and the P_2 equation eliminates T; multiplying the $\mathbf{e}_\theta$ equation by r gives the evolution of h:

$$2\ddot{r}-r\dot{\theta}^2=-g-\mu g\frac{\dot{r}}{\sqrt{\dot{r}^2+r^2\dot{\theta}^2}},\qquad \frac{d}{dt}\left(r^2\dot{\theta}\right)=-\mu g\,\frac{r^2\dot{\theta}}{\sqrt{\dot{r}^2+r^2\dot{\theta}^2}}$$

- Angular momentum and energy are no longer conserved (friction dissipates energy), so the problem cannot be reduced to the quadratures of (c).

PHASE 4

Interpretation & Checks

- **Tension at the turning points:** $T(a)=mg\left(\frac{1}{2}+\frac{4}{3}\right)=\frac{11}{6}mg$ and $T(2a)=mg\left(\frac{1}{2}+\frac{4}{24}\right)=\frac{2}{3}mg$, both positive.
- **Consistency with (a):** for $v_0^2=ag$, $h^2=a^3g$ and $\ddot{r}(a)=-\frac{g}{2}+\frac{h^2}{2a^3}=0$, so $r=a$ is an equilibrium of the radial equation, as expected for the circular case.
- **Radial period** (consequence of (c) and (d)): $\tau_r=2\sqrt{3/g}\int_a^{2a} r\,dr/\sqrt{P(r)}$. With $x=r/a$, $r\,dr=a^2x\,dx$ and $\sqrt{P}=a^{3/2}\sqrt{(x-1)(2-x)(3x+2)}$, so $\tau_r=2\sqrt{3a/g}\int_1^2 x\,dx/\sqrt{(x-1)(2-x)(3x+2)}$. Then with $x=\frac{3}{2}-\frac{1}{2}\cos s$ ($x=1\to s=0$, $x=2\to s=\pi$), $(x-1)(2-x)=\frac{1}{4}\sin^2 s$ and $dx=\frac{1}{2}\sin s\,ds$, so $\tau_r=2\sqrt{\frac{3a}{g}}\displaystyle\int_0^{\pi}\frac{x(s)\,ds}{\sqrt{3x(s)+2}}\approx 6.342\sqrt{\frac{a}{g}}$.
- **Dimensions:** $[T]=[m][g]=\mathrm{N}$ and $[v_0]=\sqrt{[a][g]}=\mathrm{m/s}$.
- **Feasibility:** the outer turning point $r_{\max}=2a$ puts P_2 at the hole with zero velocity ($\ell_2=0$), the limit of the constraint $r\le 2a$. The tension there, $T(2a)=\frac{2}{3}mg>0$, shows that the cord is still taut, so the motion is admissible for the whole cycle.
- **Check against the official key:** `Problems.pdf` has no entry for Problem 43 in its solution keys (the tension in (e) is given in the statement and is derived above). All results are therefore verified by independent derivation and by numerical integration.

- **Quadratures** (Python, DOP853 on the equations of 3.3 with $a = g = 1$): from $r = a$ to $r = 1.5a$ the integral for t gives $1.361538\sqrt{a/g}$ and the integral for θ gives 1.635065 rad, both equal to the values of the direct integration to 10^{-10}.
- **Symbolic check** (sympy): $\dot{r}^2 = \dfrac{g\,(2a - r)(r - a)(2a + 3r)}{3r^2}$ with roots $r = a$ and $r = 2a$; the tension $m(g + \ddot{r})$ equals $mg\left(\frac{1}{2} + \frac{4a^3}{3r^3}\right)$ identically.
- **Numerical check** (Python, DOP853 on $\ddot{r} = -g/2 + 4a^3 g/(3r^3)$ with $a = g = m = 1$, $r(0) = 1, \dot{r}(0) = 0$): minimum and maximum radius 1.0000 and 1.9991 over the sampled steps (the turning point 2.0000 is reached between samples), period 6.34243 (the quadrature gives 6.34243), and the energy residual stays below 5×10^{-12}.

Source: Problems.pdf (PDF page 30); no key available